

Project Synopsis

for

Design and Development of Identity Security Framework for EdTech

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I. Title of the Project

Design and Development of Identity Security Framework for EdTech

II. Statement of the Problem

Educational institutions handle vast amounts of sensitive data, including student academic records, faculty research, and institutional intelligence. Traditional learning management systems (LMS) rely on vulnerable password-based authentication mechanisms, leaving them susceptible to phishing, credential stuffing, and injection attacks. The problem is to engineer a Zero-Trust login and identity management system for the EdTech sector, utilizing Python, Flask, SQLite, and WebAuthn, that thoroughly demonstrates modern authentication, strict payload sanitization, and role-based authorization.

III. Why this particular topic chosen?

The EdTech sector is increasingly targeted by cyber threats due to the high value of scholarly data and personal identifiable information (PII). A standard username and password are no longer adequate defense mechanisms. This topic was chosen to demonstrate how modern security concepts—specifically passwordless Passkey authentication (WebAuthn), automated Intrusion Prevention Systems (IPS), and multi-tiered role isolation (Students vs. Teachers vs. Admins)—can be integrated into an educational portal to eliminate traditional threat vectors.

IV. Objective and Scope

- **Objective:** To develop a highly secure, high-fidelity web application portal for academic users using Python, Flask, SQLite, and the WebAuthn standard.
- **Scope:** Zero-Trust user registration

- WebAuthn/Passkey integration (passwordless authentication)
- Global IPS payload sanitization (SQLi/XSS detection)
- SQLite session management
- Specific access control matrices for Student, Faculty, and Admin roles
- Replacement of legacy passwords with biometric Passkeys
- Enforcement of strict quarantine protocols for malicious network activity

V. Methodology

An incremental, security-first development approach was utilized:

1. **Database Schema & Routing:** Establishing the foundational architecture.
2. **Hybrid Authentication:** Implementing TOTP and WebAuthn to replace traditional passwords.
3. **Intrusion Prevention System (IPS):** Binding an overarching IPS to all server requests.
4. **Cyber-Glass UI & AI Tutor (Orion):** Integrating a seamless, premium user experience.

VI. Process Description

- **Authentication Module:** Handles secure registration and passwordless login using FIDO2/WebAuthn standard (Passkeys) and fallback TOTP mechanisms.
- **Session Management:** Secure, encrypted Flask-Login sessions track operative states, terminating upon anomaly detection.
- **Authorization Module (The "Nexus"):** Restricts access based on hierarchical roles: Student (academic data), Teacher (class data), and Admin (system audit and clearance).
- **Intrusion Prevention System (IPS):** A globally scoped monitoring engine that scans all inbound traffic for SQLi and XSS payloads,

triggering an automatic IP quarantine (403 Forbidden) upon threat detection.

VII. Resources and Limitations

- **Hardware:** Standard development workstations and mobile devices equipped with biometric sensors (for Passkey testing).
- **Software:** Python, Flask, Flask-Login, SQLite, WebAuthn libraries, PyOTP, and Tailwind CSS.
- **Limitations:** The framework operates as an isolated software prototype and currently does not integrate directly with legacy hardware architectures, enterprise Active Directory distributions, or physical campus access control hardware.

VIII. Testing Technologies used

- **Black Box Testing:** Verification of the registration flow, IP clearance requests via the Orion AI, and role-based dashboard rendering.
- **White Box Testing:** Verification of the IPS Regex signature matching engine and cryptographic WebAuthn payload validations.
- **Security Penetration Testing:** Deliberate injection of XSS (<script>) and SQLi (' OR 1=1--) payloads to validate the automated IP quarantine mechanism and 403 lockout screens.

IX. Conclusion

The EdTech Identity Security Framework successfully proves that modern, enterprise-grade security protocols (Zero-Trust, WebAuthn, and IPS) can be effectively married to academic portals. By discarding traditional passwords and actively neutralizing hostile payloads, this framework sets a new baseline for protecting student and faculty integrations in modern learning environments.



